

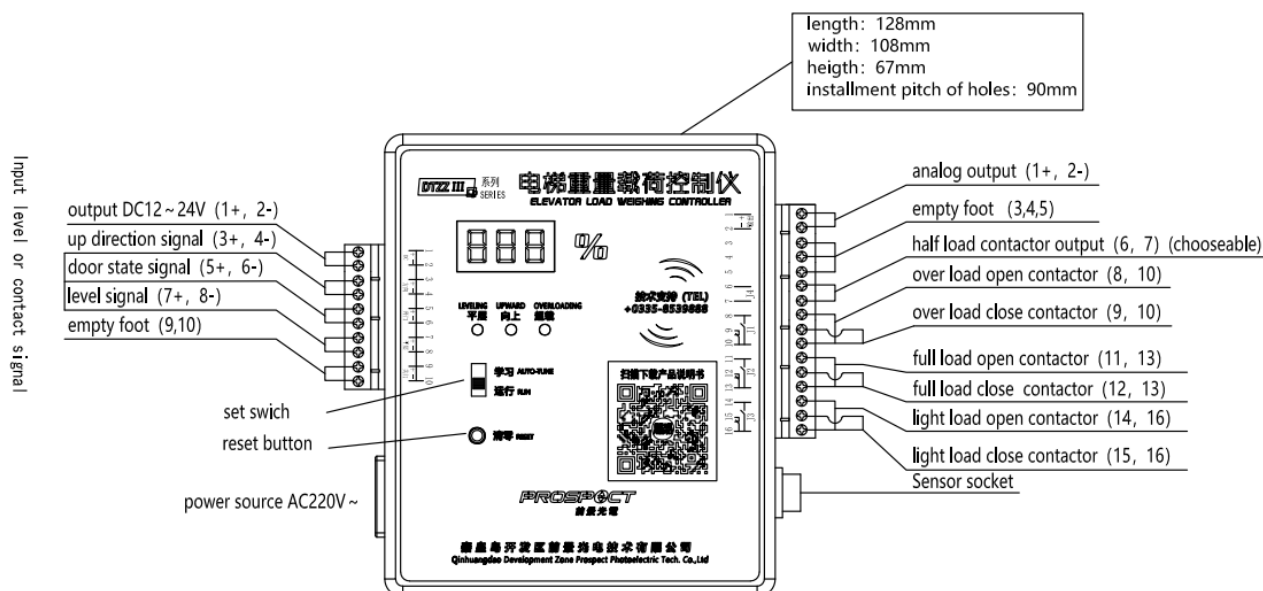
DTZZIII-DK100 Series Elevator Weight-Load Indicating Controller

Debugging Instructions v1.1.2.1.2

1 Technical parameters and types

Type	DK100	DK101	DK102	DK103
Main feature	1.No analog output 2.over load 、 full load、 light load	1.analog output is 0～10mA 2.over load、 full load、 light load	1.analog output is -10～10V 2.2.over load、 full load、 light load	1.No analog output 2.over load 、 full load
Technology parameter	Power Source		AC220（-15～+10%） V	
	Sensitivity		<0.1 %	
	Control return difference		<1 %	
	Working temperature		－10 ～ 60℃	
	Power consumption of instrument		<10 W	
	Overload capacity of sensor		150 %	
	Output contactor capacity		0.3A/DC24V； 0.3A/AC220V	
Product common	1.Product power source is AC220V； 2.Product of the input signal connection mode are the same, as shown in the second； 3.Product overload, full load, light load relay contacts are one normally open, one normally closed； 4.Product installation are all the same, as shown in the three or four。			
Note:	The half load of the product is optional, and the relay contact is normally open by default. If there is no special requirement, this function is not available.			

2 Control instrument appearance size and terminal function

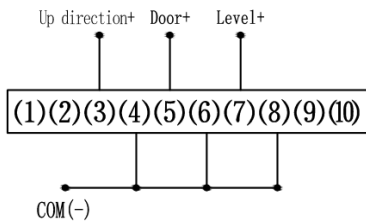


(Diagram text annotation for the terminal serial number in parentheses)

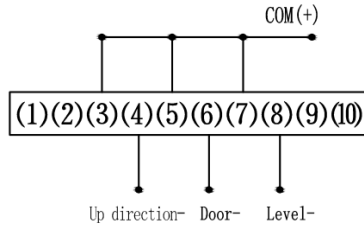
Control instrument appearance size Picture

Note:

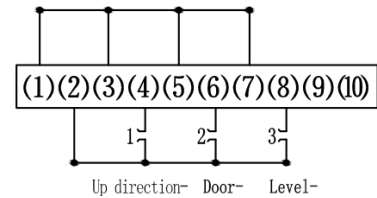
2.1 Each input signal can be a contact signal or a level signal. The wiring method is shown in the figure below. In the contact mode, when the elevator goes up, the contact 1 is closed; when the elevator opens the door, the contact 2 is closed; when the elevator is on a leveling floor, the contact 3 is closed.



Common “-” connection



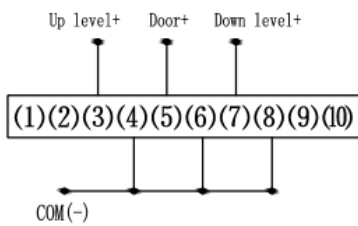
Common “+” connection



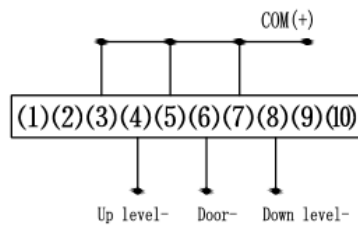
Contactor connection

Terminal wiring diagram

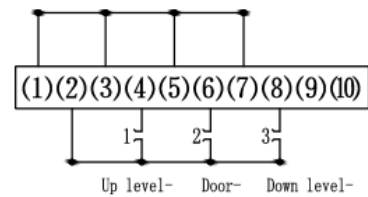
2.2 If the input signal of the controller needs to be connected to the double leveling floor, the wiring method is as shown in the figure below.



Common “-” connection



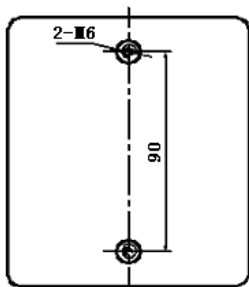
Common “+” connection



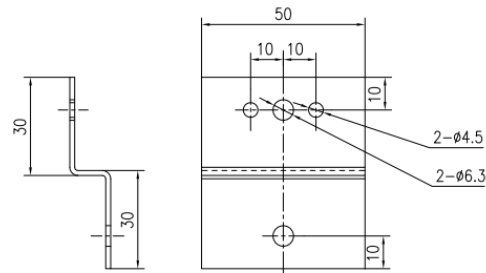
Contactor connection

Terminal wiring diagram

3 The installation size and control instrument



a、Control cabinet install

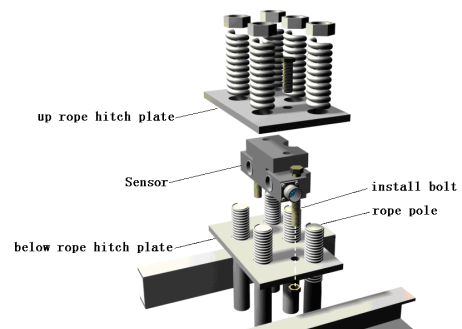
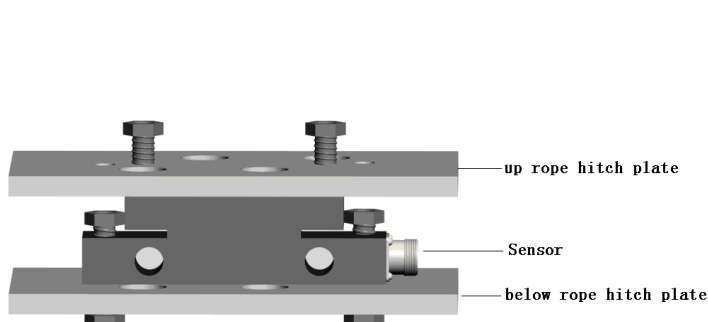


b、Director install fittings size

Director install size picture

If the controller is installed directly in the elevator control cabinet, can use the M6 bolts, according to the figure a direct control devices installed inside the control cabinet; If need to install in other places, you can install the two attachments (figure b) fixed at the back of the chassis (figure a two M6 thread), then install the control instrument accessories to a fixed location.

4 Sensor install picture



Sensor install picture

Note:

- 4.1 Additional rope head plate thickness of 20 ~ 25 mm, outside dimensions beyond hole outside around 30 mm.
- 4.2 Sensors placed between two rope head plate, connected to the top and bottom line head plate respectively, as shown above.
- 4.3 Sensors connected to the rope on the head plate bolt screw in sensor part cannot beyond sensor junction surface under the surface.
- 4.4 Rope rod must avoid contact with the sensor head.
- 4.5 Rope head must be avoided by its rope head plate hole wall contact.

5 Initial review

5.1 Power on

Make the“set”switch on running, with power on, the controller will hear one long sound,three-digit display 888. Then display --1. Open the door, three-digit will all display, and will increase with the load of the elevator increasing.

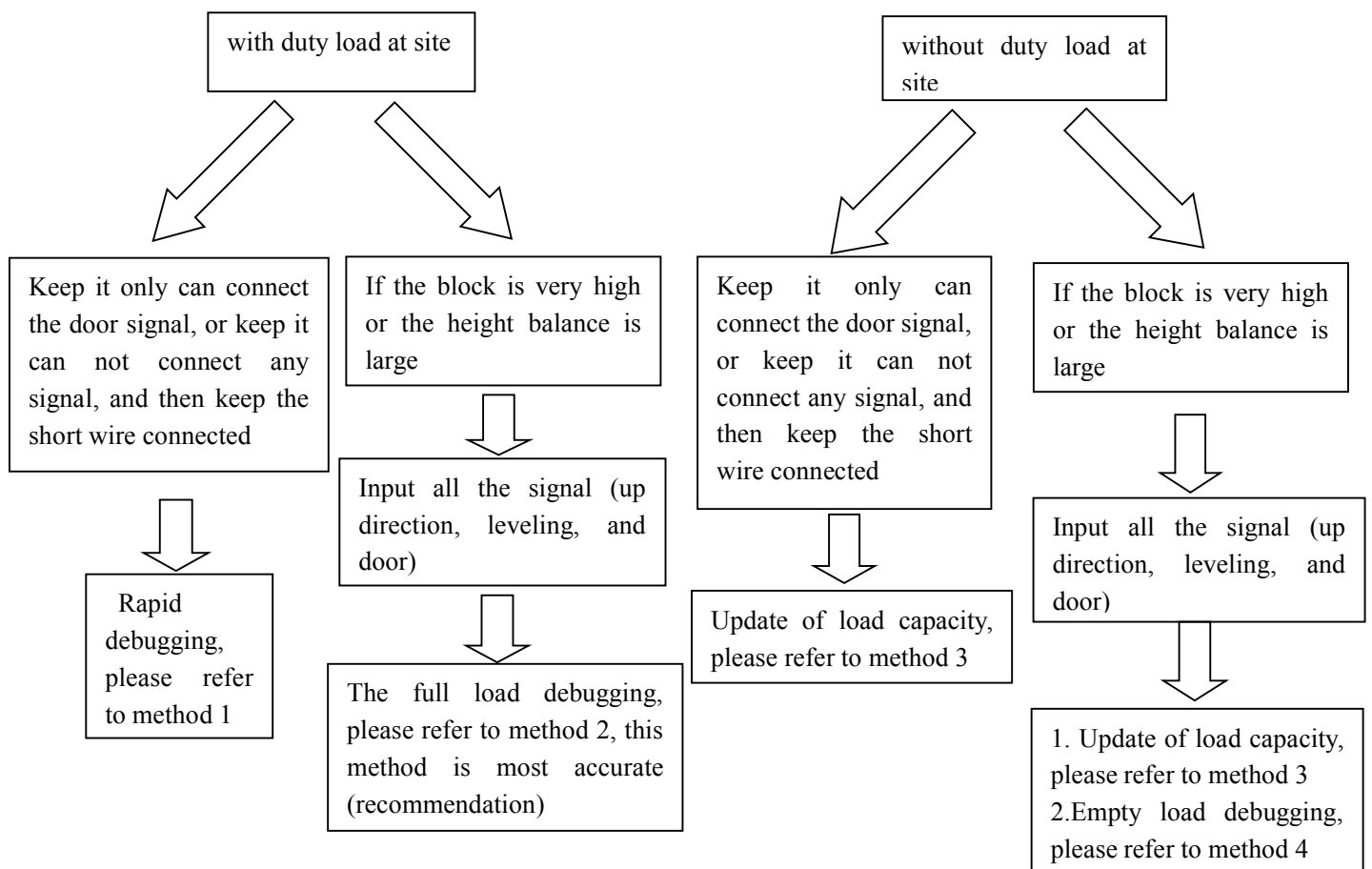
5.2 Reset

Make the“set”switch on running;

Power on and preheating for 30 seconds;

Let elevator empty and stay on the first floor, keep the door open. Press the reset button three times in three seconds continuously, at the same time you will hear three short sounds, and then three long sounds, three-digit display 000. Now the reset (get rid of the tare) is finished.

6 Debugging (according to the site, choose one method)



6.1 Method 1 Rapid debugging

6.1.1 Connect the door signal, and open the door.

6.1.2 After the initial test, keep the elevator empty and stay on the floor on which the dead weight of the car is most light, then reset it.

6.1.3 After reset finished, keep the elevator stay on the floor on which the dead weight of car is most heaviest, increase the load in the car to 100% of duty load, press the reset switch for more than 5 seconds, when hear a long sound, then the three-digit display 099 or 100, release the switch;

6.1.4 Make the controller power off and then power on, the debugging is finished.

6.2 Method 2 Full load debugging

6.2.1 After reset(get rid of the tare), keep the door open, make the load in the car to 100% of duty load, press the reset switch for more than 5 seconds, when hear a long sound, then the three-digit display 099 or 100, release the switch;

6.2.2 Press the set switch up to learning position, you will heard 3 long sounds and 1 short sound. Close the door, make the elevator stay on every floor and open the door from the first floor to the top floor one by one.(stop on every floor, the controller will learning and give a sound)

6.2.3 When reach the top floor, press the reset switch 5 times in 5 seconds continuously, at the same time you will hear 5 gently sounds, and then 5 long sounds. Now make the elevator stay on every floor and open the door from top floor to first floor one by one(stop on every floor, the controller will learning and give a sound).

6.2.4 After reach the first floor, keep the door open, press the set switch down to running position. The instrument goes into normal operation, and the three-digit display 100.

6.2.5 Make the controller power off and then power on, the debugging is finished.

6.3 Method 3 Update of load capacity

6.3.1 When the upper left corner of the instrument panel is marked as "DTZZ-III series", as shown in the figure on the right:



6.3.1.1 The elevator stops at the ground floor and opens the door, clear the empty load (tare weight)

6.3.1.2 Load $\geq 20\%$ of the load.

6.3.1.3 Power off the controller, press the reset button, and turn on the power again, turn the setting switch up to the position of <Learning>. After hearing three long beeps, press the reset button once, accompanied by a light beep.

6.3.1.4 Calculate the theoretical set value of the load according to the recorded loading load. Theoretical set value = loading load $\div$ rated load $\times 100\%$.

6.3.1.5 Press the reset button to adjust the displayed original set value to the calculated theoretical set value. (The set value cycles between 000-100).

6.3.1.6 Turn the setting switch back to the <Run> position, and then power on the controller again after power off, and the debugging is completed.

6.3.1.7 Example: The rated load of the elevator is 1000kg. When the elevator adds a load of 200kg, the theoretical set value of the device should be $200 \div 1000 \times 100\% = 20$. If the displayed value is 23%, through the reset button, manually adjust the display of the digital tube to 20.

6.3.2 When the upper left corner of the instrument panel is marked as "DTZZ-III", as shown in the figure on the right:



6.3.2.1 The elevator stops at the ground floor and opens the door, clear the empty load (tare weight).

6.3.2.2 Load more than 20% of the load and record the actual display value of the digital tube.

6.3.2.3 After the controller is powered off and re-energized, turn the setting switch up to the position of <Learning>. After hearing two long beeps, press the reset button once, accompanied by a light beep. Record the original set value of the three-digit load displayed by the device at this time.

6.3.2.4 Calculate the theoretical set value of the load amount based on the actual display value of the recorded digital tube, the theoretical display value and the original set value of the load amount, the theoretical set value = the actual display value $\times$ the original set value $\div$ the theoretical display value.

6.3.2.5 Press the reset button to adjust the displayed original set value to the calculated theoretical set value. (The set value cycles between 020-255).

6.3.2.6 Turn the setting switch back to the <Run> position, and then power on the controller again after power off, and the debugging is completed.

6.3.2.7 Example: The rated load of the elevator is 1000kg. When the elevator adds a load of 200kg, The theoretical display value of the device should be 20% (loading value $200 \div$ rated load $1000 \times 100\%$), while the actual display value is 23%, the original set value is 100, the theoretical set value is $23 \times 100 \div 20 = 115$, manually adjust the theoretical set value to 115.

6.4 Method 4 Empty load debugging

6.4.1 First please reset (get rid of the tare) the instrument.

6.4.2 Press the set switch up to learning position, you will heard 3 long sounds and 1 short sound. Close the door, make the elevator stay on every floor and open the door from the first floor to the top floor one by one.(stop on every floor, the controller will learning and give a sound)

6.4.3 When reach the top floor, press the reset switch 5 times in 5 seconds continuously, at the same time you will hear 5 gently sounds, and then 5 long sounds. Now make the elevator stay on every floor and open the door from top floor to first floor one by one(stop on every floor, the controller will learning and give a sound).

6.4.4 After reach the first floor, keep the door open, press the set switch down to running position. The instrument goes into normal operation, and the three-digit display 000.Now the debugging is finished.

7 Set the light load point

When leaving factory, the default setting of light point is 10%. If you need reset the light point, please operate as follow:

When the instrument is in normal working state (elevator is no-load), press the reset button to hear a response, after hearing a short beep within 2S, turn the setting switch up immediately, release the reset button, and hear three long beeps(Note: When the upper left corner of the instrument panel is marked as "DTZZ-III series", it will be three long sounds, and when the upper left corner of the instrument panel is marked as "DTZZ-III", it will be one long sound), enter the adjustment state, press the reset button to adjust the light load point. The displayed value will cycle between 002-050. After adjusting to the desired light load point, pull down the setting switch to exit the state of setting the light load point.

8 Fault handling

In normal operation, if the instrument display 666, please check whether the sensor connect the controller well.

Note: when the elevator opens the door, the instrument test the duty load, and output the contactor signal and electricity signal. When the elevator closes the door, the electricity signal will not change.

If any question, or any faulty with controller, please contact us in time.

When the product is scrapped, the relevant disposal regulations related to the list of hazardous wastes must be strictly implemented.

Technical service: 0335-8539888/8539818

Manufacturer: Qinhuangdao Development Zone Prospect Photoelectric Tech. Co., Ltd.

**Address: No. 3 Jinghe path, Middle Huashan Road, Qinhuangdao Development Zone,
Hebei Province.**

Post Code: 066004